

Submit 8 problems total from the union of *LADW* Ch 2.4-8 (copied below) and the supplementary problems. Please label all *LADW* problems by “Chapter.Section.Exercise,” e.g. 1.3.4. Label the supplementary problems as they’re labeled on the handout. Please make a note about any collaboration or use of outside resources. Please put a “G” in front of the four problems you would like graded for correctness. Additionally, put an “F” in front of any of the problems graded for completion on which you would like written feedback from the grader.

Suggested *LADW* problems: 2.5.2, 2.5.4, 2.6.2, 2.7.4, 2.8.3

Section 2.4

Exercises.

X

4.1. Find the inverse of the matrices

$$\begin{pmatrix} 1 & 2 & 1 \\ 3 & 7 & 3 \\ 2 & 3 & 4 \end{pmatrix}, \quad \begin{pmatrix} 1 & -1 & 2 \\ 1 & 1 & -2 \\ 1 & 1 & 4 \end{pmatrix}.$$

Show all steps

Section 2.5

Exercises.

5.1. True or false:

- Every vector space that is generated by a finite set has a basis;
- Every vector space has a (finite) basis;
- A vector space cannot have more than one basis;
- If a vector space has a finite basis, then the number of vectors in every basis is the same.
- The dimension of $\mathbb{P}_n$ is n ;
- The dimension on $M_{m \times n}$ is $m + n$;
- If vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ generate (span) the vector space V , then every vector in V can be written as a linear combination of vector $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ in only one way;
- Every subspace of a finite-dimensional space is finite-dimensional;
- If V is a vector space having dimension n , then V has exactly one subspace of dimension 0 and exactly one subspace of dimension n .

5.2. Prove that if V is a vector space having dimension n , then a system of vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ in V is linearly independent if and only if it spans V .

5.3. Prove that a linearly independent system of vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ in a vector space V is a basis if and only if $n = \dim V$.

5.4. (An old problem revisited: now this problem should be easy) Is it possible that vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ are linearly dependent, but the vectors $\mathbf{w}_1 = \mathbf{v}_1 + \mathbf{v}_2$, $\mathbf{w}_2 = \mathbf{v}_2 + \mathbf{v}_3$ and $\mathbf{w}_3 = \mathbf{v}_3 + \mathbf{v}_1$ are linearly independent? **Hint:** What dimension can the subspace $\text{span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$ have?

5.5. Let vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$ be a basis in V . Show that $\mathbf{u} + \mathbf{v} + \mathbf{w}$, $\mathbf{v} + \mathbf{w}$, $\mathbf{w}$ is also a basis in V .

5.6. Consider in the space $\mathbb{R}^5$ vectors $\mathbf{v}_1 = (2, -1, 1, 5, -3)^T$, $\mathbf{v}_2 = (3, -2, 0, 0, 0)^T$, $\mathbf{v}_3 = (1, 1, 50, -921, 0)^T$.

- Prove that these vectors are linearly independent.
- Complete this system of vectors to a basis.

If you do part b) first you can do everything without any computations.

Section 2.6

Exercises.

6.1. True or false

- Any system of linear equations has at least one solution;
- Any system of linear equations has at most one solution;
- Any homogeneous system of linear equations has at least one solution;
- Any system of n linear equations in n unknowns has at least one solution;
- Any system of n linear equations in n unknowns has at most one solution;
- If the homogeneous system corresponding to a given system of a linear equations has a solution, then the given system has a solution;
- If the coefficient matrix of a homogeneous system of n linear equations in n unknowns is invertible, then the system has no non-zero solution;
- The solution set of any system of m equations in n unknowns is a subspace in $\mathbb{R}^n$;
- The solution set of any homogeneous system of m equations in n unknowns is a subspace in $\mathbb{R}^n$.

6.2. Find a 2×3 system (2 equations with 3 unknowns) such that its general solution has a form

$$\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + s \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \quad s \in \mathbb{R}.$$

Section 2.7

Exercises.

7.1. True or false:

- The rank of a matrix is equal to the number of its non-zero columns;
- The $m \times n$ zero matrix is the only $m \times n$ matrix having rank 0;
- Elementary row operations preserve rank;
- Elementary column operations do not necessarily preserve rank;
- The rank of a matrix is equal to the maximum number of linearly independent columns in the matrix;
- The rank of a matrix is equal to the maximum number of linearly independent rows in the matrix;
- The rank of an $n \times n$ matrix is at most n ;
- An $n \times n$ matrix having rank n is invertible.

7.2. A 54×37 matrix has rank 31. What are dimensions of all 4 fundamental subspaces?

X

7.3. Compute rank and find bases of all four fundamental subspaces for the matrices

$$\begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & 2 & 3 & 1 & 1 \\ 1 & 4 & 0 & 1 & 2 \\ 0 & 2 & -3 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

7.4. Prove that if $A : X \rightarrow Y$ and V is a subspace of X then $\dim AV \leq \text{rank } A$. (AV here means the subspace V transformed by the transformation A , i.e. any vector in AV can be represented as $A\mathbf{v}$, $\mathbf{v} \in V$). Deduce from here that $\text{rank}(AB) \leq \text{rank } A$.

Remark: Here one can use the fact that if $V \subset W$ then $\dim V \leq \dim W$. Do you understand why is it true?

7.5. Prove that if $A : X \rightarrow Y$ and V is a subspace of X then $\dim AV \leq \dim V$. Deduce from here that $\text{rank}(AB) \leq \text{rank } B$.

7.6. Prove that if the product AB of two $n \times n$ matrices is invertible, then both A and B are invertible. Even if you know about determinants, do not use them, we did not cover them yet. **Hint:** use previous 2 problems.

7.7. Prove that if $A\mathbf{x} = \mathbf{0}$ has unique solution, then the equation $A^T \mathbf{x} = \mathbf{b}$ has a solution for every right side $\mathbf{b}$.

Hint: count pivots

7.8. Write a matrix with the required property, or explain why no such matrix exists

- Column space contains $(1, 0, 0)^T$, $(0, 0, 1)^T$, row space contains $(1, 1)^T$, $(1, 2)^T$;
- Column space is spanned by $(1, 1, 1)^T$, nullspace is spanned by $(1, 2, 3)^T$;
- Column space is $\mathbb{R}^4$, row space is $\mathbb{R}^3$.

Hint: Check first if the dimensions add up.

7.9. If A has the same four fundamental subspaces as B , does $A = B$?

7.10. Complete the rows of a matrix

$$\begin{pmatrix} e^3 & 3 & 4 & 0 & -\pi & 6 & -2 \\ 0 & 0 & 2 & -1 & \pi^e & 1 & 1 \\ 0 & 0 & 0 & 0 & 3 & -3 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

to a basis in $\mathbb{R}^7$.

X **7.11.** For a matrix

$$\begin{pmatrix} 1 & 2 & -1 & 2 & 3 \\ 2 & 2 & 1 & 5 & 5 \\ 3 & 6 & -3 & 0 & 24 \\ -1 & -4 & 4 & -7 & 11 \end{pmatrix}$$

find bases in its column and row spaces.

7.12. For the matrix in the previous problem, complete the basis in the row space to a basis in $\mathbb{R}^5$

7.13. For the matrix

$$A = \begin{pmatrix} 1 & i \\ i & -1 \end{pmatrix}$$

compute $\text{Ran } A$ and $\text{Ker } A$. What can you say about relation between these subspaces?

7.14. Is it possible that for a real matrix A that $\text{Ran } A = \text{Ker } A^T$? Is it possible for a complex A ?

7.15. Complete the vectors $(1, 2, -1, 2, 3)^T$, $(2, 2, 1, 5, 5)^T$, $(-1, -4, 4, 7, -11)^T$ to a basis in $\mathbb{R}^5$.

Section 2.8

Exercises.

8.1. True or false

- Every change of coordinate matrix is square;
- Every change of coordinate matrix is invertible;
- The matrices A and B are called similar if $B = Q^T A Q$ for some matrix Q ;
- The matrices A and B are called similar if $B = Q^{-1} A Q$ for some matrix Q ;
- Similar matrices do not need to be square.

8.2. Consider the system of vectors

$$(1, 2, 1, 1)^T, \quad (0, 1, 3, 1)^T, \quad (0, 3, 2, 0)^T, \quad (0, 1, 0, 0)^T.$$

- Prove that it is a basis in $\mathbb{F}^4$. Try to do minimal amount of computations.
- Find the change of coordinate matrix that changes the coordinates in this basis to the standard coordinates in $\mathbb{F}^4$ (i.e. to the coordinates in the standard basis $\mathbf{e}_1, \dots, \mathbf{e}_4$).

8.3. Find the change of coordinates matrix that changes the coordinates in the basis $1, 1+t$ in $\mathbb{P}_1$ to the coordinates in the basis $1-t, 2t$.

8.4. Let T be the linear operator in $\mathbb{F}^2$ defined (in the standard coordinates) by

$$T \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3x+y \\ x-2y \end{pmatrix}$$

Find the matrix of T in the standard basis and in the basis

$$(1, 1)^T, \quad (1, 2)^T.$$

8.5. Prove, that if A and B are similar matrices then $\text{trace } A = \text{trace } B$. **Hint:** recall how $\text{trace}(XY)$ and $\text{trace}(YX)$ are related.

8.6. Are the matrices

$$\begin{pmatrix} 1 & 3 \\ 2 & 2 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 0 & 2 \\ 4 & 2 \end{pmatrix}$$

similar? Justify.

Supplementary problems for HW3

I. i) Give three examples of matrices in $M_{2 \times 2}^{\mathbb{R}}$ that are equal to their own inverses.

ii) Give three examples of matrices in $M_{2 \times 2}^{\mathbb{Z}/5\mathbb{Z}}$ that are equal to their own inverses.

II. i) Characterize all matrices in $M_{2 \times 2}^{\mathbb{R}}$ that are equal to their own inverses.

ii) Characterize all matrices in $M_{2 \times 2}^{\mathbb{Z}/5\mathbb{Z}}$ that are equal to their own inverses. How many are there?

III. Always sometimes, or never? (If sometimes, give an example each way. If always or never, explain why.)

i) If A and B are square matrices and $AB = B$, then A is an identity matrix.

ii) The rank of a matrix M is equal the rank of M^T .

iii) The nullity of a matrix M is equal to the nullity of M^T .

iv) A square matrix is similar to an identity matrix.

v) A square matrix is row equivalent to an identity matrix.

vi) A square matrix is similar to an identity matrix.

vii) An invertible matrix is row equivalent to an identity matrix.

viii) if $x_1 = 2$, $x_2 = 5$, $x_3 = 8$ is a solution to a homogeneous system of linear equation over $\mathbb{R}$, then the solution is unique.

IV. Work out an LU factorization and the LDU factorization of each of the following real matrices:

$$\begin{pmatrix} 2 & 2 \\ 6 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 & 6 \\ 1 & 3 & 5 \\ 4 & 2 & 1 \end{pmatrix}$$

V. Give an example of system of linear equations that has exactly 25 solutions, or explain why this is not possible.